Mini Project Report

on

Medical Insurance Claim Fraud Detection: Multi-Paradigm Machine Learning, Deep Learning, Agentic XAI, and Verified Dual-Path Deployment

Submitted by

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Certificate

This is to certify that the project, entitled **Medical Insurance Claim Fraud Detection: Multi-Paradigm Machine Learning, Deep Learning, Agentic XAI, and Verified Dual-Path Deployment**, is a bonafide record of the Mini Project coursework presented by the students whose names are given below during Academic Year 2025–2026 in partial fulfilment of the requirements of the degree of Bachelor of Technology in Computer Science and Engineering (Data Science & AI cohort).

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Abstract

Healthcare insurance fraud imposes substantial financial losses on insurers and beneficiaries while demanding low false-alarm triage and transparent audit justification for human investigators. On a verified corpus of 4,500 medical insurance claims (270 fraudulent, 6.00% prevalence), this project presents a unified, leakage-governed fraud intelligence platform uniting three complementary AI approaches—Classical Machine Learning (5 algorithms), PyTorch Deep Learning (4 supervised architectures plus 1 unsupervised Autoencoder ablation), and Agentic AI with Explainability (XAI) and Gemini conversational assistance—deployed across both a three-tier React/Node/FastAPI application (Track A, 29 features) and a controlled browser-native benchmark lab (Track B, 4 experimental protocols over 32 runs). A rigorous dataset audit exposed two critical pitfalls: an upstream *Cluster* variable acting as a leaky label proxy (concentrating 267 of 270 frauds in Cluster 1) and an amount boundary separating all positives at ClaimAmount ≥ 7,465.07. Under the leakage-free *Intake* contract (28 features excluding Cluster) and strict validation-F1 thresholding, Random Forest and XGBoost achieve 1.0000 test Average Precision (PR-AUC), with Random Forest attaining F1 = 0.9873 (0 false positives, 1 missed fraud) and maintaining F1 = 1.0000 under chronological future testing (≥ 2024-03-11). Among deep learners on the 29-feature contract, TabTransformer leads with 0.9911 PR-AUC (F1 = 0.8989), followed by CNN1D (0.9882), ResidualMLP (0.9561), and MLP (0.9538), whereas an unsupervised Autoencoder trained solely on legitimate claims yields a documented negative result (0.0523 PR-AUC, 0.0299 F1). Exported browser weights match Python reference execution across 10,800 score evaluations within 4.42e-7 maximum absolute difference with zero decision flips.

Contents & Index of Artifacts

1. Introduction **1**

1.1 Problem Statement and Research Objectives 1

1.2 Project Contributions and Team Ownership 1

2. Literature Survey and Related Work **2**

2.1 Classical Ensembles & Systematic Reviews in Healthcare Claims 2

2.2 Deep Learning for Tabular Data & Autoencoder Anomaly Detection 2

2.3 Imbalanced Evaluation, Leakage, XAI & Agentic Workflows 2

3. Data, Problem Formulation, Innovation & Implementation **2**

3.1 Audited Corpus Characteristics & Leakage Hazards 2

3.2 Mathematical Problem Formulation & Experimental Contracts 3

3.3 Innovation & Three-Approach Dual-Path Implementation 4

4. Experimental Results and Discussions **5**

4.1 Track A Full 10-Model Benchmark & Autoencoder Negative Study 5

4.2 Track B Sensitivity Ablations, Forensics & Parity Verification 6

5. Conclusion and Future Work **7**

References **8**

List of Figures

Figure 1: Exploratory Data Audit (Upstream Cluster Leakage & Intake RF Feature Importance) . . . . p. 3

Figure 2: Unified End-to-End Dual-Path System Architecture (Approaches 1–3, Track A & Track B) . . . . p. 4

Figure 3: Approach 3 Agentic Runtime Status vs. 6-Role Design & XAI Scaffold . . . . p. 5

Figure 4: Track A Unified 10-Model Test Benchmark (Classical ML vs. Supervised DL vs. Autoencoder) . . . . p. 6

Figure 5: Track B Four-Experiment Sensitivity & Chronological Ablation Summary . . . . p. 7

Figure 6: Track B Champion Confusion Matrix, Row 1124 Audit & Python-to-Browser Parity Ledger . . . . p. 7

List of Tables

Table 1: Controlled Feature & Chronological Split Contracts (Track A vs. Track B) . . . . p. 3

Table 2: Track A Held-Out Test Performance Across All 10 Implemented ML & DL Models . . . . p. 5

Table 3: Track B Test PR-AUC (and Validation-F1 Tuned F1) Across Four Protocols . . . . p. 6

1. Introduction

Healthcare insurance billing systems process high-volume outpatient, inpatient, routine, and emergency claims daily, creating severe exposure to upcoding, duplicate billing, phantom procedures, and eligibility abuse [11, 13]. Automating fraud screening requires resolving five coupled technical tensions rather than merely fitting an off-the-shelf binary classifier: (1) severe positive-class scarcity (6.00% fraud prevalence in our corpus), (2) vulnerability to upstream feature leakage and single-feature shortcuts [17, 18], (3) divergence between tree ensembles and deep neural networks on tabular claims [8], (4) temporal drift under future claim rollouts, and (5) Special Investigation Unit (SIU) demands for human-understandable attribution [19].

1.1 Problem Statement and Research Objectives

- **Leakage & Distributional Governance:** Audit every column in *Health Insurance Fraud Claims.xlsx* for identifier memorization, upstream clustering proxies (*Cluster*), and single-variable shortcuts (*ClaimAmount*), defining explicit ablation contracts.
- **Multi-Paradigm Benchmarking:** Implement and evaluate 10 models across Classical ML (Approach 1), PyTorch Deep Learning (Approach 2), and Agentic AI / LLM triage (Approach 3).
- **Imbalance-Aware Evaluation:** Prioritize PR-AUC / Average Precision over optimistic accuracy or ROC-AUC [16], tune operating thresholds strictly on validation folds, and compute 1,000-replicate bootstrap 95% confidence intervals.
- **Dual-Path Delivery & Numerical Parity:** Deliver both a live three-tier React/Node/FastAPI application serving all 10 checkpoints and Gemini chat (Track A) and a zero-server pure-JavaScript Claim Fraud Lab verified within 1e-6 numerical tolerance (Track B).

1.2 Project Contributions and Team Ownership

Team Member	Roll No	Primary Ownership & Engineering Scope
B. Varshith	23BDS011	Approach 3: Agentic workflow UI (6-role dashboard), Express AgenticEnsemble-v1 handler, Gemini 2.0 Flash /api/chat co-pilot, SHA-256 XAI utility scaffold.
J. Ganesh Kumar Reddy	23BDS024	Approach 1 & Research Lab: Dataset audit, leakage/shortcut isolation, 5 classical ML models, submission/ 32-run ablation benchmark, pure-JS weight exporter & parity tests.
M. Jagadeeswar	23BDS033	Approach 2 & Path A Full-Stack: 5 PyTorch DL architectures (MLP, ResidualMLP, CNN1D, TabTransformer, Autoencoder), FastAPI ml-service (:8000), Node/React ML-DL integration.

2. Literature Survey and Related Work

2.1 Classical Ensembles & Systematic Reviews in Healthcare Claims

du Preez et al. [13] systematically reviewed 137 healthcare claim fraud studies, concluding that Random Forests [1] and gradient boosted decision trees (XGBoost [2], LightGBM [3]) consistently dominate tabular fraud tasks while criticizing widespread omission of temporal splits and audit-cost evaluation. Bauder and Khoshgoftaar [11] confirmed similar gains on Medicare Part B billing over linear and kernel baselines [4], while Kapadiya et al. [14] highlighted the architectural integration of predictive ML with tamper-evident cryptographic audit logs.

2.2 Deep Learning for Tabular Data & Autoencoder Anomaly Detection

Johnson and Khoshgoftaar [12] demonstrated that class-imbalanced Medicare neural networks require weighted cross-entropy or sampling (e.g., SMOTE [15]) to avoid minority collapse. Recent tabular architectures add residual skip blocks [5, 7] and categorical self-attention via TabTransformer [6]. Nevertheless, Grinsztajn et al. [8] proved across 45 medium-sized (~10K-sample) datasets that axis-aligned tree splits handle irregular target boundaries and uninformative covariates more reliably than smooth neural hyperplanes. Meanwhile, Sakurada and Yairi [9] and Chalapathy and Chawla [10] warned that reconstruction autoencoders struggle when anomalies reside inside the convex interior of normal data covariates.

2.3 Imbalanced Evaluation, Leakage, XAI & Agentic Workflows

Saito and Rehmsmeier [16] proved PR-AUC is strictly more informative than ROC-AUC under high class skew because ROC dilutes false alarms across massive negatives. Kaufman et al. [17] and Geirhos et al. [18] formalized data leakage and shortcut learning traps, while SHAP [19] and Retrieval-Augmented Generation (RAG) [20] establish the foundation for explainable, evidence-backed agentic SIU workflows.

3. Data, Problem Formulation, Innovation & Implementation

3.1 Audited Corpus Characteristics & Leakage Hazards

All experiments originate from *Health Insurance Fraud Claims.xlsx* (4,500 rows, 19 columns, 0 missing, 0 duplicates, dates 2022-07-09 to 2024-07-08, 270 frauds = 6.00%). Six identifier or near-unique fields (ClaimID, PatientID, ProviderID: 4,500 unique; DiagnosisCode & ProcedureCode: 4,495 unique; ProviderLocation: 3,876 unique) were excluded to block record memorization. Exploratory audit uncovered two severe hazards shown in Figure 1: (1) **Upstream Cluster Leakage:** 267 of 270 frauds (98.89%) reside in Cluster 1 (24.27% rate vs. 0.00% in Clusters 0/2 and 0.26% in Cluster 3); (2) **Billing Amount Boundary:** all frauds satisfy ClaimAmount ≥ 7,465.07 (mean 8,733.57 vs. legitimate range 100.12–9,997.20).

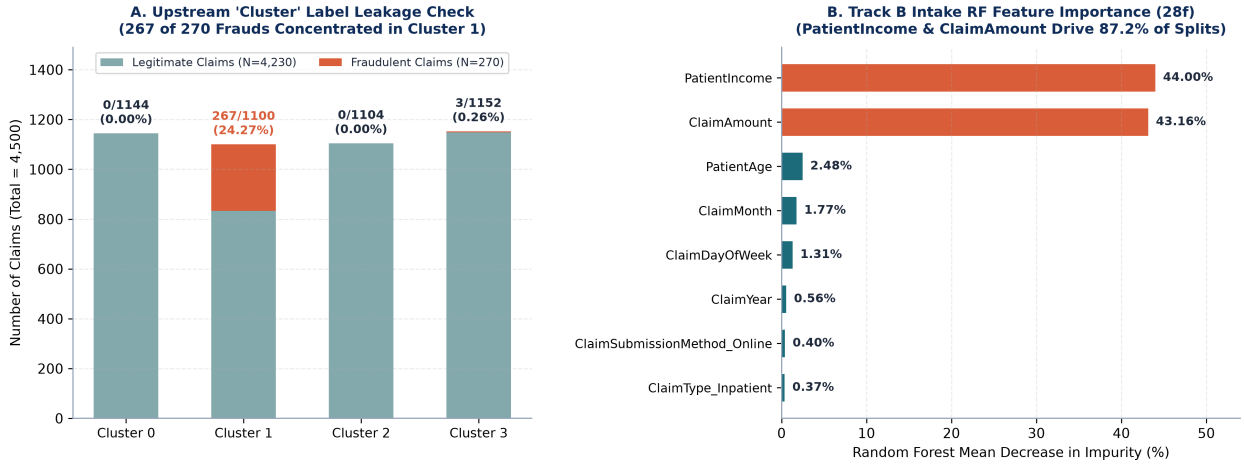


Figure 1: Dataset Leakage & Shortcut Audit — (A) Cluster 1 fraud concentration (98.89%); (B) Intake RF split importance (87.2% driven by PatientIncome & ClaimAmount).

3.2 Mathematical Problem Formulation & Experimental Contracts

A training-only *ColumnTransformer* fits *StandardScaler* on continuous attributes (including *ClaimYear*, *ClaimMonth*, and *ClaimDayOfWeek* parsed from *ClaimDate*) alongside *OneHotEncoder* (with unknown-category safe handling) on 7 administrative and clinical categoricals. Supervised PyTorch models minimize imbalance-weighted binary cross-entropy:

$$L_{wBCE}(\theta) = -\frac{1}{N_{tr}} \sum \left[w_+ \cdot y_i \cdot \log(\sigma(z_i)) + (1 - y_i) \cdot \log(1 - \sigma(z_i)) \right]$$
$$\text{where } w_+ = N_- / N_+ \approx 15.67$$

The Autoencoder minimizes legitimate-only MSE reconstruction error $\|x - D(E(x))\|^2$ with threshold selected unsupervisedly at the 95th validation percentile (0.128588). Table 1 summarizes our four controlled protocols isolating legacy serving compatibility (Track A) from deployment realism (Track B).

Protocol / Contract	Dim	Split Rule	Train / Val / Test (Frauds)	Scientific Purpose
Baseline (Track A & B)	29	Stratified Seed 42	3,150 (189) / 675 (41) / 675 (40)	Full-stack 10-model API contract retaining Cluster.
Intake (Track B)	28	Stratified Seed 42	3,150 (189) / 675 (41) / 675 (40)	Deployment-realistic intake excluding Cluster.
No-Amount (Track B)	27	Stratified Seed 42	3,150 (189) / 675 (41) / 675 (40)	Shortcut stress test removing Cluster & ClaimAmount.
Temporal Intake (Track B)	28	Chronological Date	3,152 (199) / 677 (34) / 671 (37)	Forward-in-time test (Train <= 2023-11-21; Test >= 2024-03-11).

Table 1: Controlled feature and chronological split contracts separating Track A API alignment from Track B leakage governance.

3.3 Innovation & Three-Approach Dual-Path Implementation

As depicted in Figure 2 and Figure 3, our engineering implementation bridges three complementary paradigms across two verified deployment paths: (1) **Approach 1 (Classical ML):** balanced LogisticRegression, RandomForest (200 trees), XGBoost (scale_pos_weight=15.67, eval_metric=aucpr), RBF SVM, and HistGradientBoostingClassifier; (2) **Approach 2 (PyTorch DL):** FraudMLP (29→64→32→1 with BatchNorm/Dropout), ResidualMLP, FraudCNN1D (1D convolutions over aligned feature vectors), TabTransformer (categorical self-attention), and legitimate-only Autoencoder; (3) **Approach 3 (Agentic AI, XAI & Co-Pilot):** Express *POST /agentic/predict* (AgenticEnsemble-v1 emitting 4 stage trace events), *POST /api/chat* powered by Google Gemini 2.0 Flash, a 6-agent React UI pipeline, and an offline XAIExplainer SHA-256 audit utility. Path A serves all 10 trained checkpoints behind FastAPI (:8000) → Express (:3000) → React (:5173), while Path B exports verified weights to zero-server JavaScript (:8080).

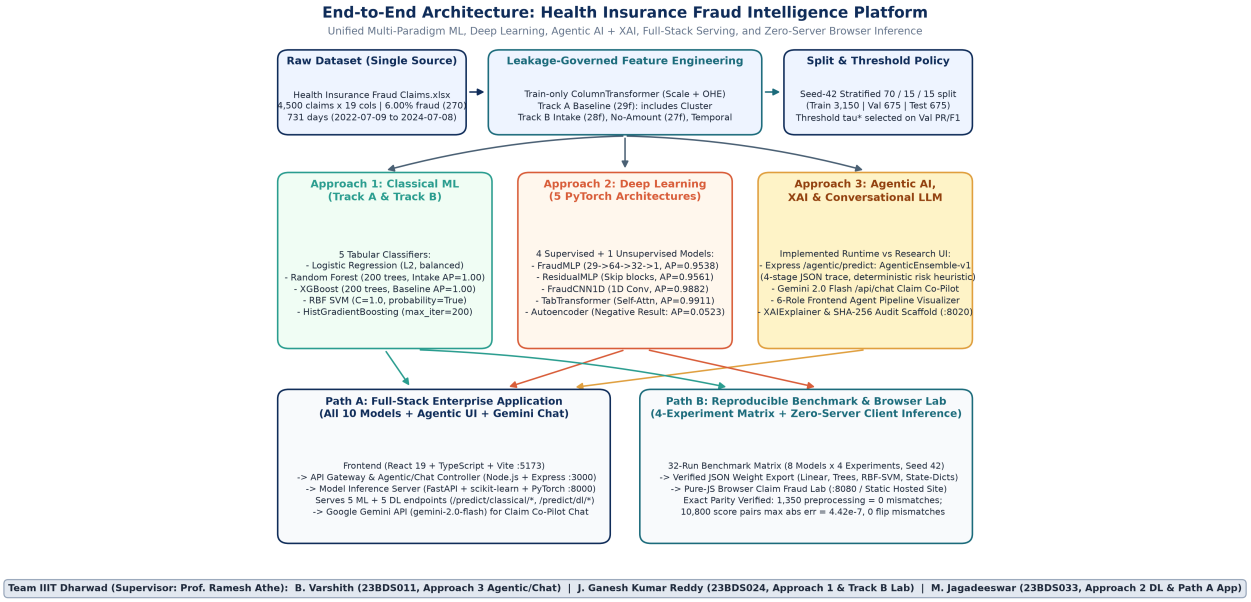


Figure 2: End-to-End Platform Architecture uniting Approaches 1–3, Full-Stack Path A (:5173/:3000/:8000), and Browser Parity Path B (:8080).

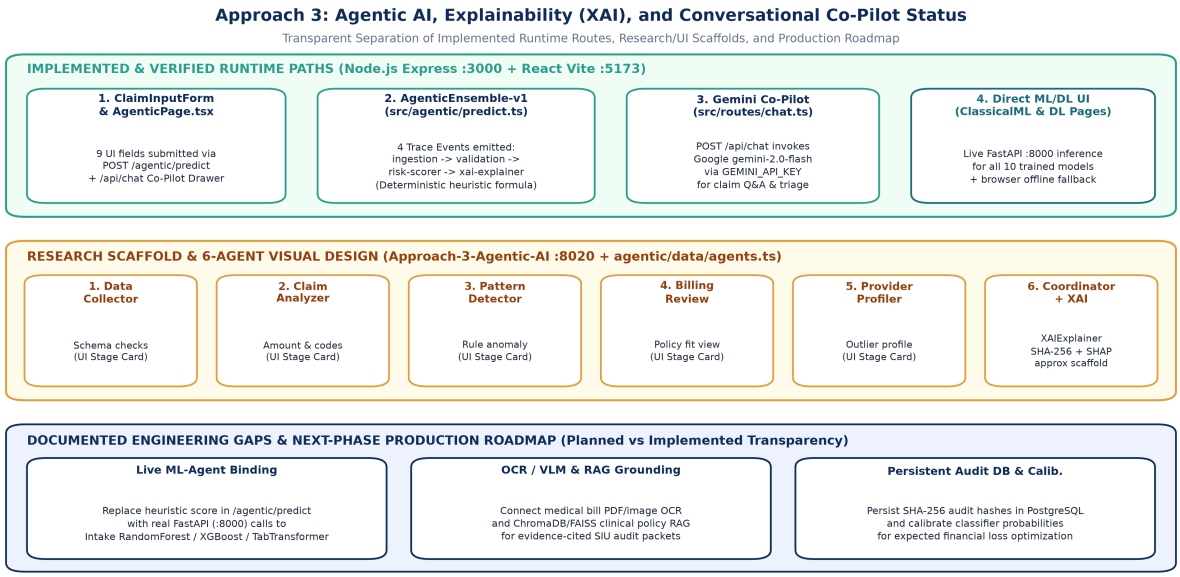


Figure 3: Approach 3 honest implementation audit separating live Express/Gemini APIs from the 6-agent UI view, XAI scaffold, and future RAG roadmap.

4. Experimental Results and Discussions

4.1 Track A Full 10-Model Benchmark & Autoencoder Negative Study

Table 2 and Figure 4 summarize held-out 675-claim test metrics across all 10 models on the 29-feature contract. XGBoost leads with 100.00% PR-AUC and 98.77% F1 (1 FP, 0 FN), closely matched by RandomForest (99.94% PR-AUC, 98.73% F1, 0 FP, 1 FN). Among deep learners, TabTransformer achieves the highest continuous ranking quality (99.11% PR-AUC, 99.94% ROC-AUC, 100.00% recall), while CNN1D reaches 90.91% F1 at the fixed 0.50 cutoff in 0.77 ms (vs. 18.65 ms for TabTransformer). Crucially, the unsupervised Autoencoder fails cleanly (5.23% PR-AUC, 2.99% F1, 46.02% ROC-AUC), empirically demonstrating that fraud examples overlapping legitimate covariate ranges cannot be separated reliably by global reconstruction loss alone.

Model	Approach / Family	Acc (%)	Prec (%)	Rec (%)	F1 (%)	ROC-AUC (%)	PR-AUC (%)
XGBoost	Approach 1 (ML)	99.85	97.56	100.00	98.77	100.00	100.00
RandomForest	Approach 1 (ML)	99.85	100.00	97.50	98.73	99.996	99.94
HistGradientBoosting	Approach 1 (ML)	99.85	100.00	97.50	98.73	98.47	97.65
TabTransformer	Approach 2 (Sup. DL)	98.67	81.63	100.00	89.89	99.94	99.11
CNN1D	Approach 2 (Sup. DL)	98.81	83.33	100.00	90.91	99.92	98.82
ResidualMLP	Approach 2 (Sup. DL)	98.52	80.00	100.00	88.89	99.75	95.61
MLP	Approach 2 (Sup. DL)	98.81	84.78	97.50	90.70	99.73	95.38
SVM (RBF)	Approach 1 (ML)	97.78	74.51	95.00	83.52	99.55	92.52
LogisticRegression	Approach 1 (ML)	96.44	62.50	100.00	76.92	99.50	91.86
Autoencoder (Neg.)	Approach 2 (Unsup. DL)	90.37	3.70	2.50	2.99	46.02	5.23

Table 2: Track A held-out test metrics across all 10 implemented ML and DL models (29 features, 40 frauds).

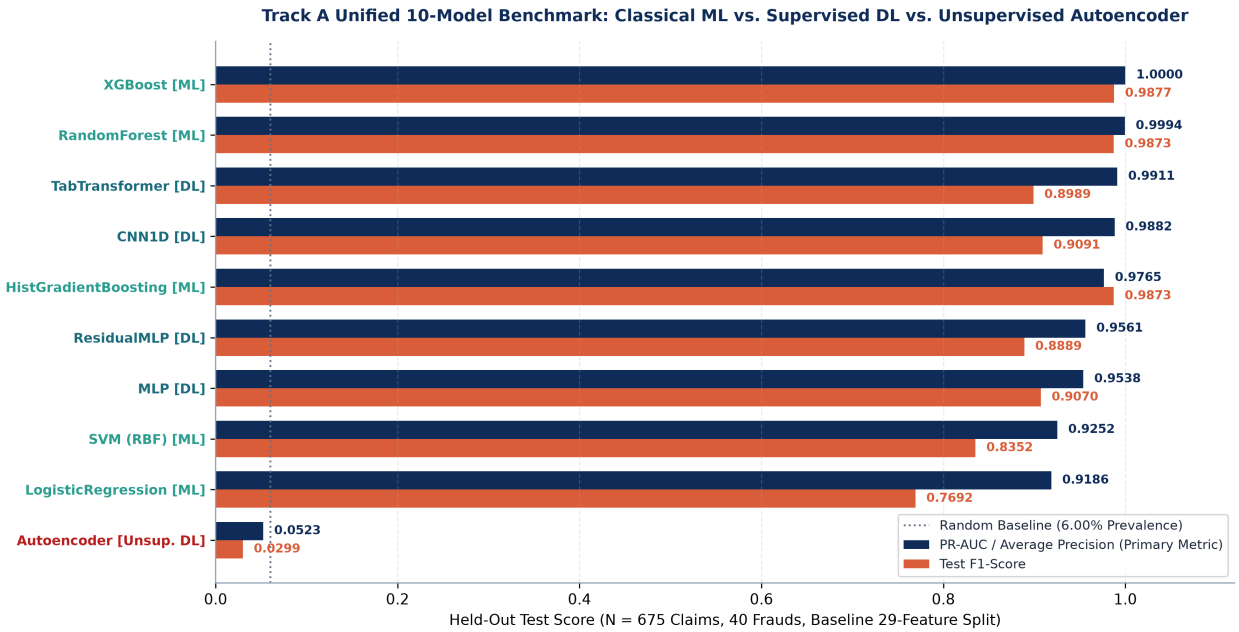


Figure 4: Track A 10-model PR-AUC vs. F1 benchmark showing ensemble dominance and the Autoencoder negative result.

4.2 Track B Sensitivity Ablations, Forensics & Parity Verification

Removing upstream *Cluster* (Table 3, Figure 5) sharply degrades non-tree models (SVM PR-AUC falls 0.9252→0.8181; ResidualMLP 0.9567→0.8765), whereas RandomForest retains 1.0000 PR-AUC and 0.9873 F1 (tau*=0.4167, 0 FP, 1 FN). Under chronological rollout (≥ 2024-03-11), all three tree models reach 1.0000 PR-AUC and 1.0000 F1. Conversely, dropping ClaimAmount in No-Amount collapses every model cleanly to 0.1869–0.2404 AP. Figure 6 inspects the single Intake RandomForest miss (Row 1124, ClaimAmount=7,465.07, lowest positive amount in the dataset), reports its 1,000-replicate bootstrap 95% F1 CI ([0.9610, 1.0000]), and confirms zero browser mismatches across 1,350 preprocessing checks and 10,800 score evaluations (max err 4.42e-7).

Model	Baseline (29f)	Intake (28f, No Cluster)	No-Amount (27f)	Temporal Intake (>=2024-03)
RandomForest	0.9994 (0.9750)	1.0000 (0.9873)	0.2066 (0.3636)	1.0000 (1.0000)
XGBoost	1.0000 (0.9877)	1.0000 (0.9877)	0.2128 (0.3524)	1.0000 (1.0000)
GradientBoosting	0.9765 (0.9873)	0.9765 (0.9873)	0.2404 (0.3636)	1.0000 (1.0000)
CNN1D	0.9882 (0.9268)	0.9667 (0.8767)	0.2287 (0.2788)	0.9172 (0.8108)
ResidualMLP	0.9567 (0.9286)	0.8765 (0.8315)	0.2246 (0.3438)	0.7979 (0.7229)
MLP	0.9538 (0.9136)	0.8866 (0.8222)	0.2303 (0.3077)	0.8581 (0.6957)
LogisticRegression	0.9186 (0.7835)	0.8525 (0.7327)	0.2318 (0.3191)	0.8799 (0.7397)
SVM (RBF)	0.9252 (0.8974)	0.8181 (0.7711)	0.1869 (0.3041)	0.8396 (0.7619)

Table 3: Track B Test PR-AUC (and validation-F1 tuned F1) across all 4 controlled scenarios (32 training runs).

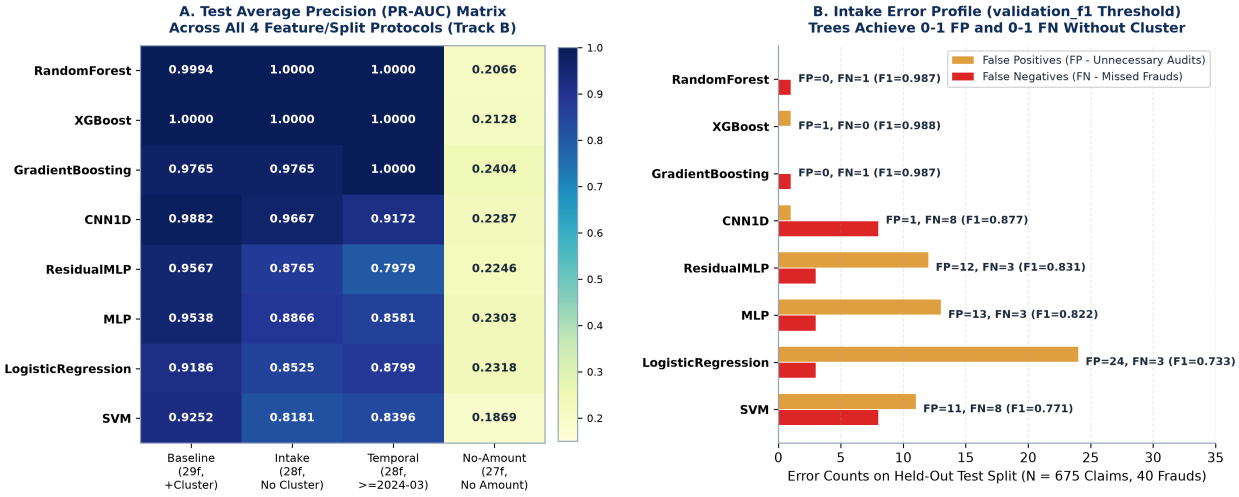


Figure 5: Track B ablation matrix (PR-AUC across 4 contracts) and Intake error counts highlighting zero false positives for RandomForest and HistGradientBoosting.

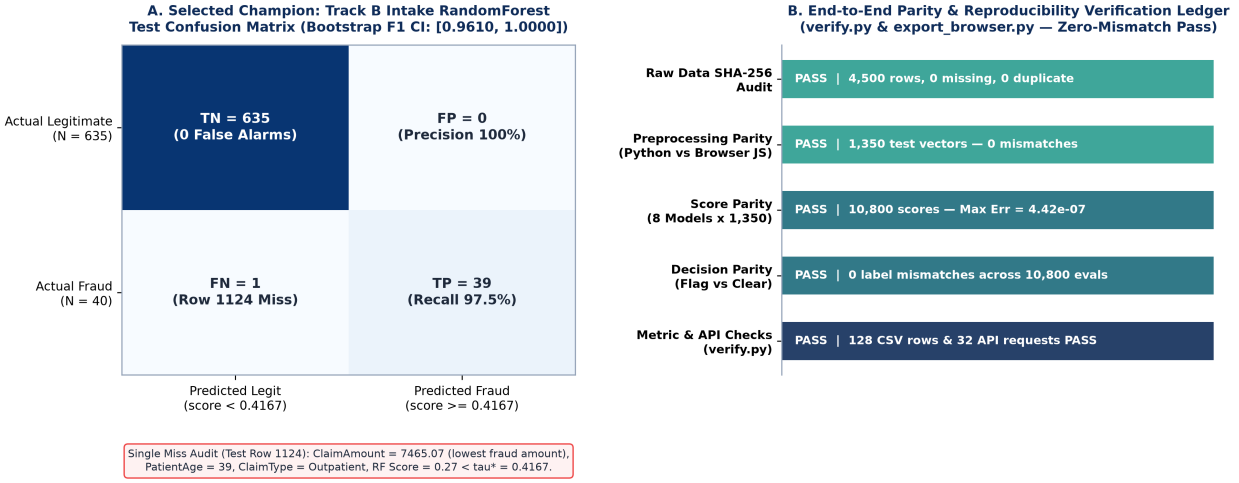


Figure 6: Track B Intake RandomForest confusion matrix (tau=0.4167), Row 1124 audit, and full Python-to-Browser verification ledger.*

5. Conclusion and Future Work

This project delivers an end-to-end, rigorously verified Medical Insurance Claim Fraud Detection system spanning Classical Machine Learning, five Deep Learning paradigms, interactive Gemini-assisted Agentic XAI, a full-stack React/Node/FastAPI application, and a numerically identical browser-native Claim Fraud Lab. Controlled ablations confirm that eliminating upstream *Cluster* leakage preserves tree ensemble superiority (1.0000 PR-AUC on both Intake and Chronological tests) while exposing neural fragility and structural dependence on *ClaimAmount*. Future work will bind Node's */agentic/predict* handler directly to calibrated FastAPI checkpoints, connect OCR/VLM bill ingestion with policy RAG vectors, persist SHA-256 SIU audit ledgers in PostgreSQL, and evaluate low-dollar upcoding detection on broader real-world claim benchmarks.

References

[1] L. Breiman, "Random Forests," Machine Learning, vol. 45, no. 1, pp. 5-32, 2001. doi:10.1023/A:1010933404324.

[2] T. Chen and C. Guestrin, "XGBoost: A Scalable Tree Boosting System," in Proc. 22nd ACM SIGKDD, 2016. doi:10.1145/2939672.2939785.

[3] G. Ke et al., "LightGBM: A Highly Efficient Gradient Boosting Decision Tree," in NeurIPS, vol. 30, pp. 3149-3157, 2017.

[4] C. Cortes and V. Vapnik, "Support-vector networks," Machine Learning, 1995. doi:10.1007/BF00994018.

[5] K. He, X. Zhang, S. Ren, and J. Sun, "Deep Residual Learning for Image Recognition," in Proc. IEEE CVPR, 2016. doi:10.1109/CVPR.2016.90.

[6] X. Huang, A. Khetan, M. Cvitkovic, and Z. Karnin, "TabTransformer: Tabular Data Modeling Using Contextual Embeddings," arXiv:2012.06678, 2020.

[7] Y. Gorishniy, I. Rubachev, V. Khrulkov, and A. Babenko, "Revisiting Deep Learning Models for Tabular Data," in NeurIPS, 2021.

[8] L. Grinsztajn, E. Oyallon, and G. Varoquaux, "Why do tree-based models still outperform deep learning on tabular data?" in NeurIPS, 2022 (arXiv:2207.08815).

[9] M. Sakurada and T. Yairi, "Anomaly Detection Using Autoencoders with Nonlinear Dimensionality Reduction," in Proc. MLSDA Workshop, 2014.

[10] R. Chalapathy and S. Chawla, "Deep Learning for Anomaly Detection: A Survey," arXiv preprint, 2019.

[11] R. A. Bauder and T. M. Khoshgoftaar, "Medicare Fraud Detection Using Machine Learning Methods," in IEEE ICMLA, 2017. doi:10.1109/ICMLA.2017.00048.

[12] J. M. Johnson and T. M. Khoshgoftaar, "Medicare fraud detection using neural networks," J. Big Data, vol. 6, art. 63, 2019. doi:10.1186/s40537-019-0225-0.

[13] A. du Preez, S. Bhattacharya, P. Beling, and E. Bowen, "Fraud detection in healthcare claims using machine learning: A systematic review," Artificial Intelligence in Medicine, vol. 160, art. 103061, 2025. doi:10.1016/j.artmed.2024.103061.

[14] K. Kapadiya et al., "Blockchain and AI-Empowered Healthcare Insurance Fraud Detection: An Analysis, Architecture, and Future Prospects," IEEE Access, vol. 10, pp. 79606-79627, 2022. doi:10.1109/ACCESS.2022.3194569.

[15] N. V. Chawla et al., "SMOTE: Synthetic Minority Over-sampling Technique," JAIR, vol. 16, pp. 321-357, 2002. doi:10.1613/jair.953.

[16] T. Saito and M. Rehmsmeier, "The Precision-Recall Plot Is More Informative than the ROC Plot When Evaluating Binary Classifiers on Imbalanced Datasets," PLOS ONE, 10(3):e0118432, 2015. doi:10.1371/journal.pone.0118432.

[17] S. Kaufman, S. Rosset, C. Perlich, and O. Stitelman, "Leakage in Data Mining: Formulation, Detection, and Avoidance," ACM TKDD, 6(4), 2012. doi:10.1145/2382577.2382579.

[18] R. Geirhos et al., "Shortcut learning in deep neural networks," Nature Machine Intelligence, 2020. doi:10.1038/s42256-020-00257-z.

[19] S. M. Lundberg and S.-I. Lee, "A Unified Approach to Interpreting Model Predictions," in NeurIPS, pp. 4765-4774, 2017.

[20] P. Lewis et al., "Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks," in NeurIPS, pp. 9459-9474, 2020 (arXiv:2005.11401).